v. Rancidity:

Rancidity is a term generally used to denote unpleasant odours and flavours in foods resulting from deterioration in the fat or oil portion of a food. The triacylglycerols in fats with low molecular mass carboxylic acids undergo oxidation very quikly when exposed to air, moisture and light or hydrolysed in the presence of bacterial lipases.

How does the food industry reduces rancidity?

- Packing foods either with naturally occurring antioxidants such as Vitamin C and vitamin E or with synthetic oxidation inhibitors such as Butylated hydroxyanisole (BHA) and Butylated hydroxytoluene (BHT).
- Vacuum packaging to avoid oxygen.
- Adding inert gas such as nitrogen to the bag to replace the oxygen.
- Packing food protected from light.
- Refrigeration will reduce the rate of most reactions that take place in rancidity.

6.4. Quantitation of fat

There are certain chemical parameters used for the characterization of fats:

i. Acid number:

Acid value (or "neutralization number" or "acid number") is the number of milligram of potassium hydroxide (KOH) that is required to neutralize one gram of fat. Thus, acid number is an indication of the quantity of free fatty acid present in the fat. The fat which has been processed and stored properly has a very low acid number.

ii. Saponification number:

The saponification number is the milligrams of KOH required to saponify 1 g of fat. Thus it provides the information about the average chain length of the fatty acids in the fat. The saponification number varies inversely with the chain length of the fatty acid. The shorter the average chain length of the fatty acid, the higher is the saponification number.

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